

SYSTEM ARCHITECTURE WITH COGNITIVE FALLBACK

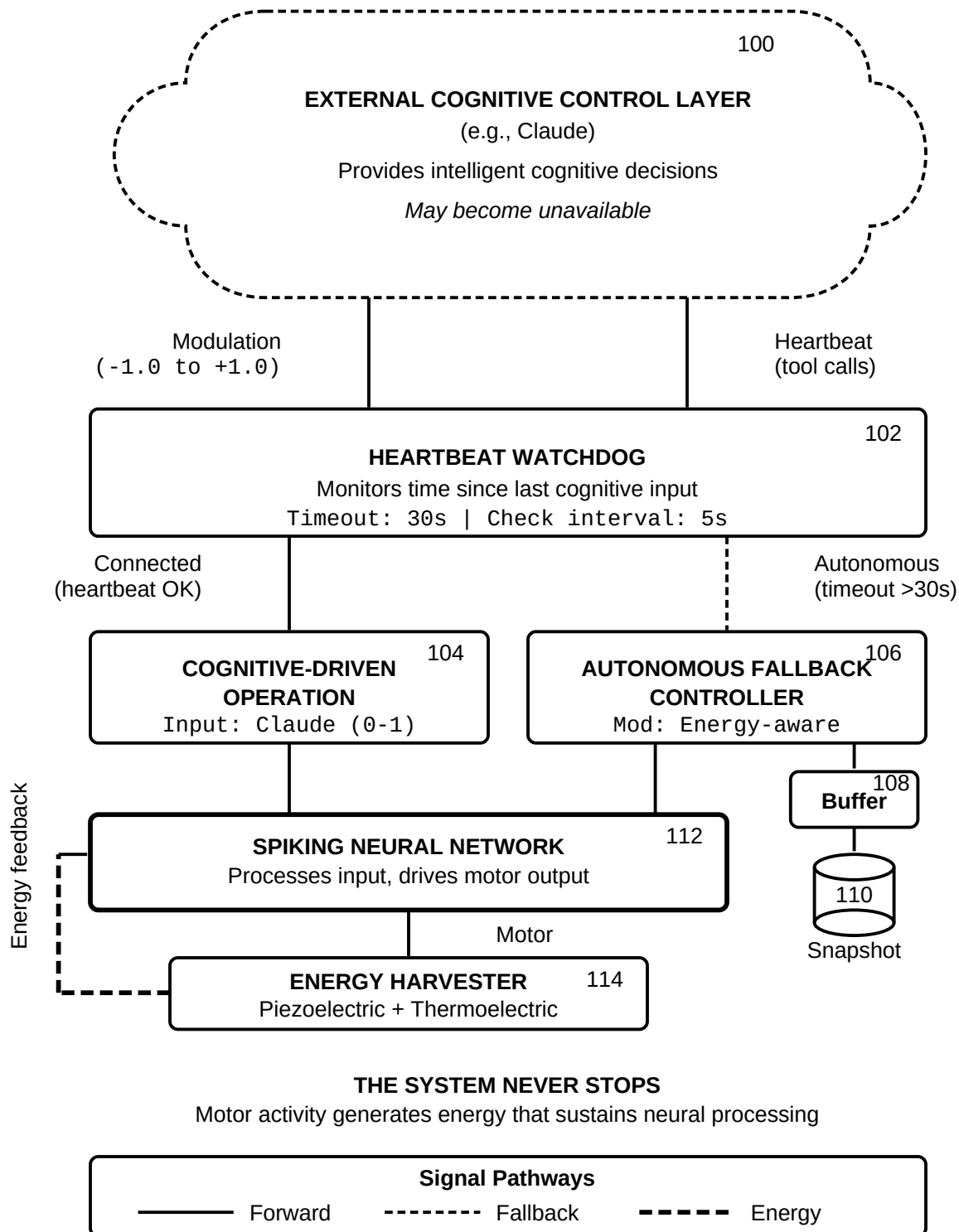
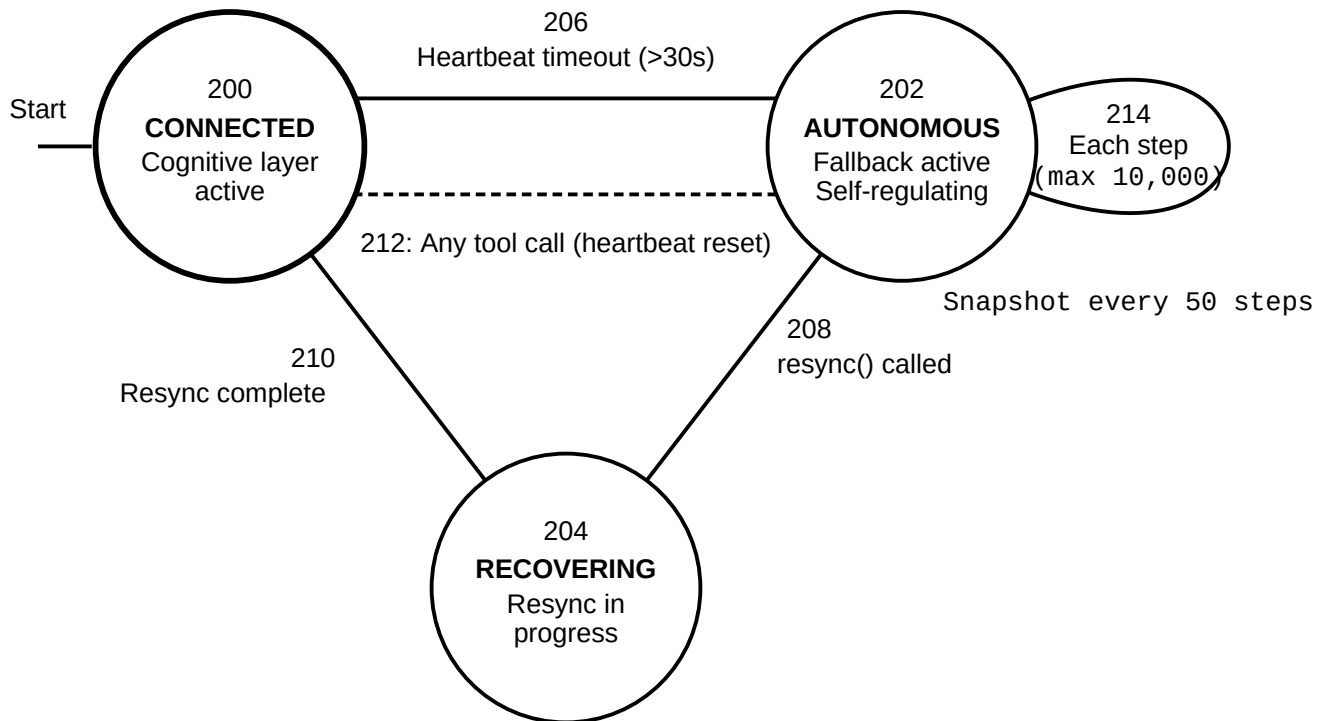


FIG. 1

STATE TRANSITION DIAGRAM



Transition Summary

206: Timeout triggers fallback

212: Direct reconnection

208: Explicit resync request

214: Autonomous step loop

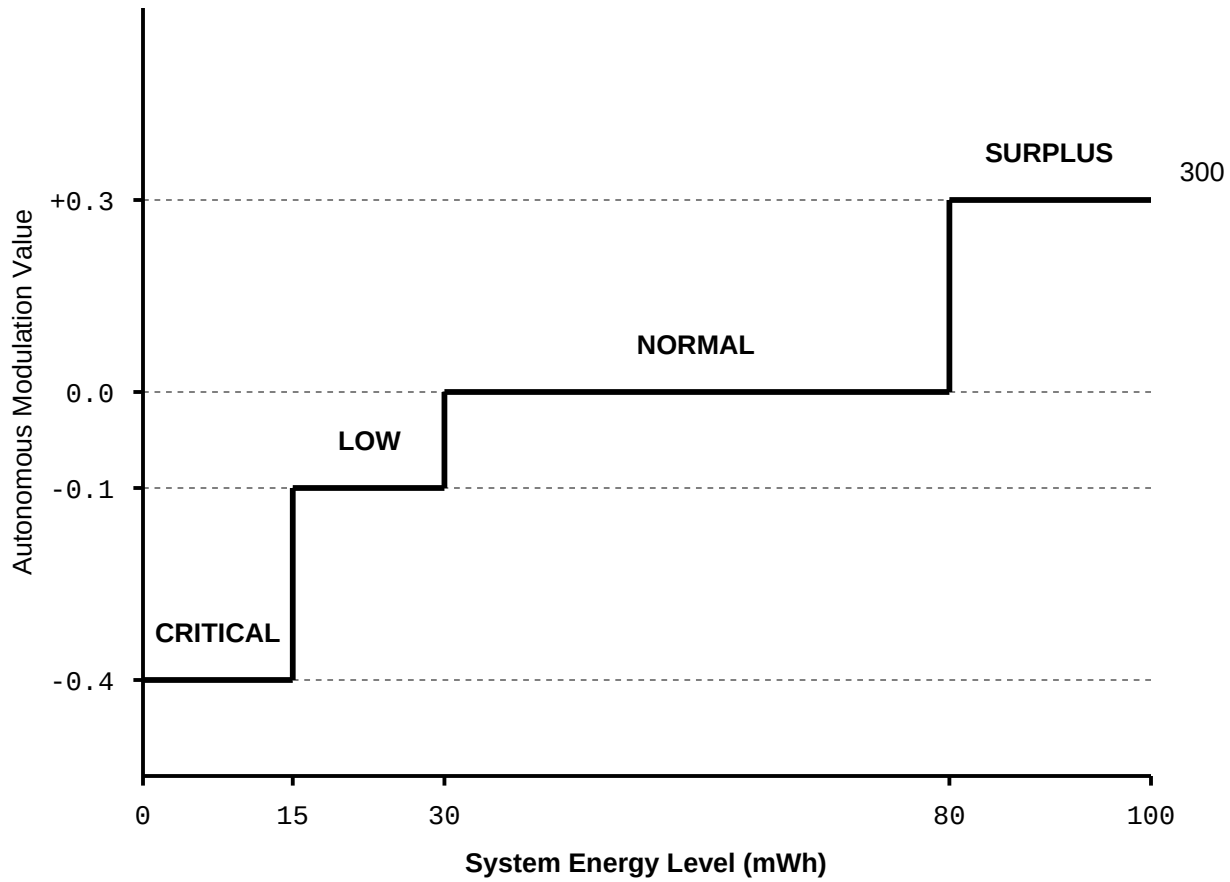
210: Resync completes

—— Initial state

----- Direct reconnect

FIG. 2

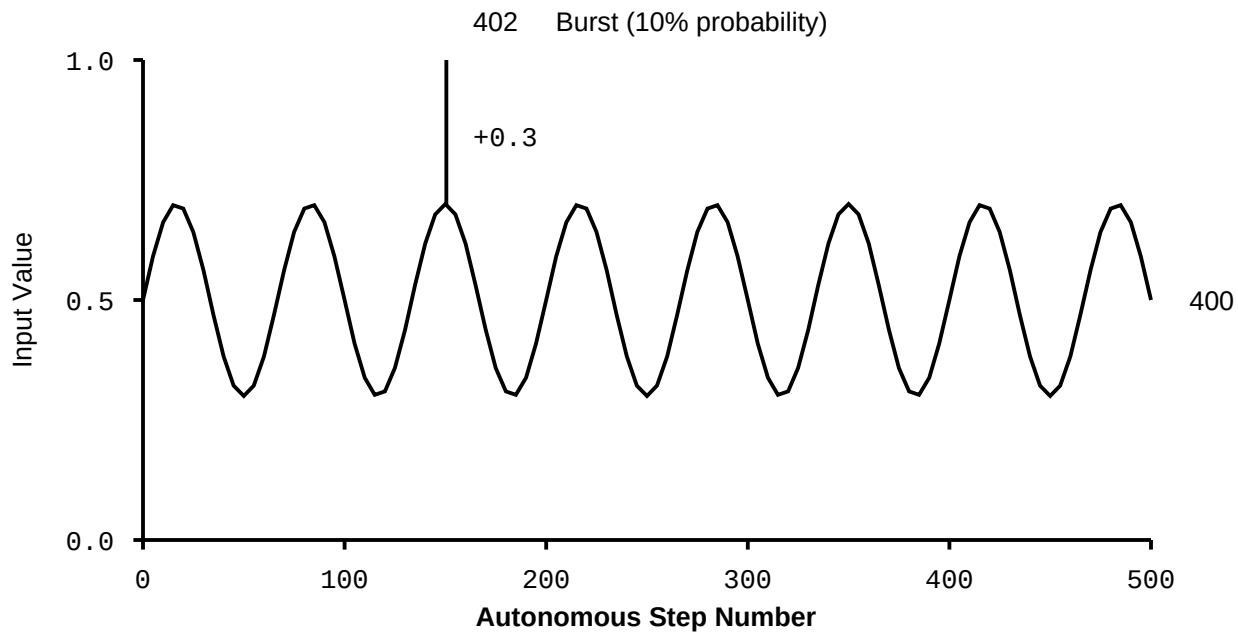
ENERGY-AWARE MODULATION CURVE



Behavioral Effects			302
< 15 mWh	mod = -0.4	Maximum conservation	
15-30 mWh	mod = -0.1	Cautious operation	
30-80 mWh	mod = 0.0	Standard processing	
> 80 mWh	mod = +0.3	Exploration enabled	

FIG. 3

AUTONOMOUS INPUT GENERATOR OUTPUT



404

Autonomous Input Formula

$$\text{circadian} = 0.5 + 0.2 \times \sin(2\pi \times t \times 3)$$

burst = 0.3 (with 10% probability per step)

Energy Recovery During Autonomous Operation

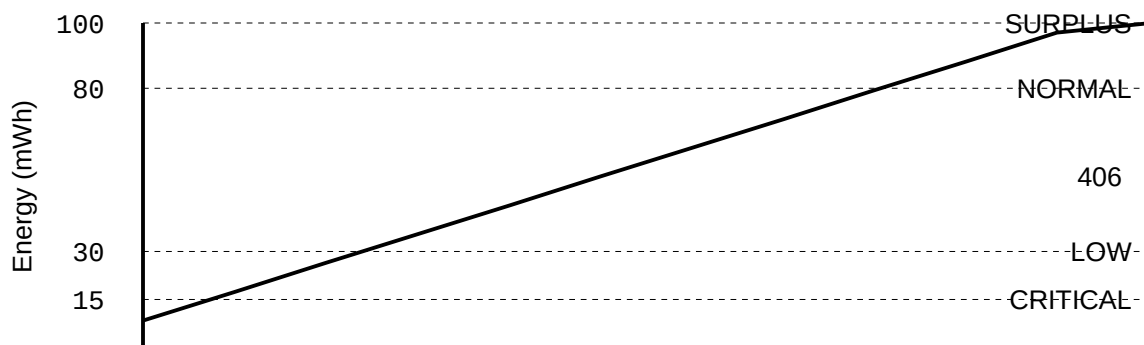


FIG. 4

RESYNCHRONIZATION PAYLOAD STRUCTURE

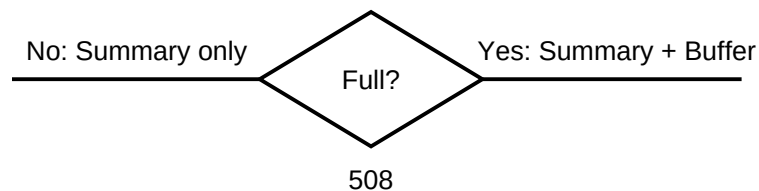
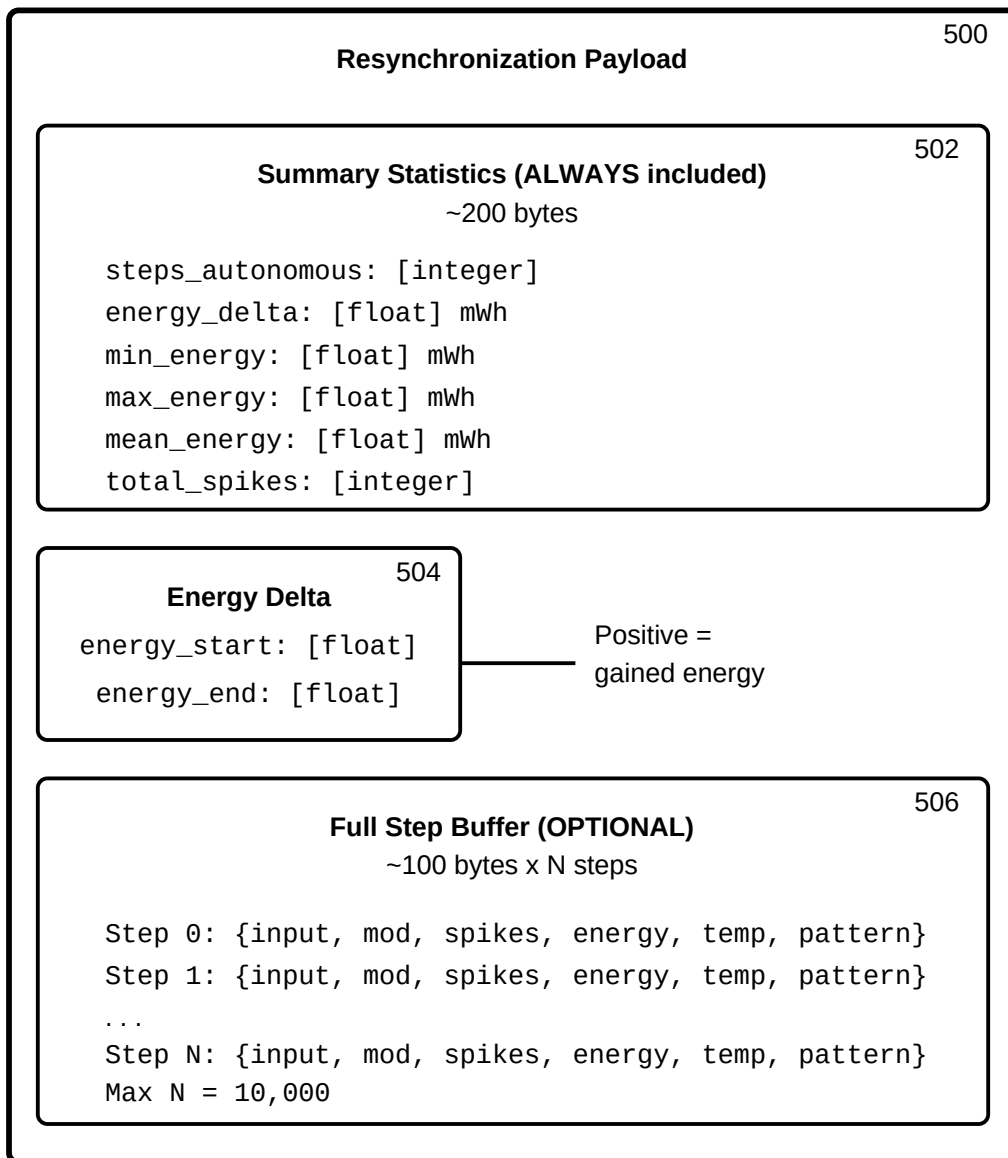
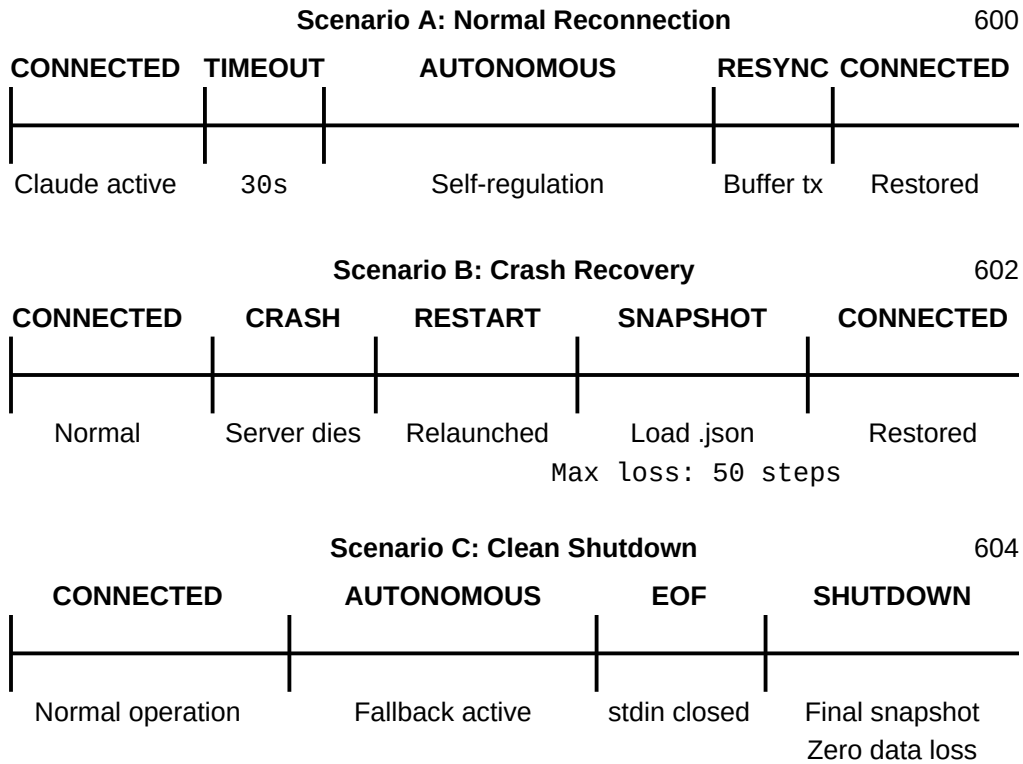


FIG. 5

RECOVERY SCENARIOS



Recovery Guarantees

606

Normal reconnection: Full continuity, resync payload transmitted

Crash recovery: Max 50 steps lost (snapshot every 50 steps)

Clean shutdown: Zero data loss (final snapshot on EOF)

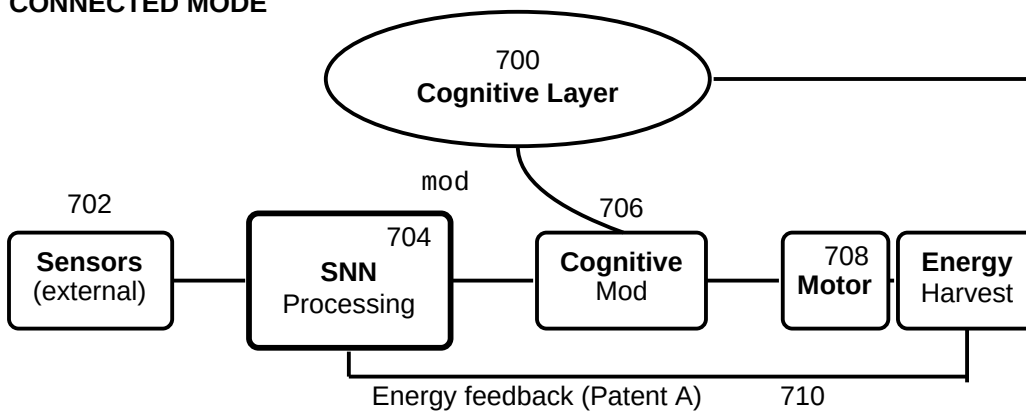
Max autonomous duration: 10,000 steps (~50 hours)

—— System actively processing

FIG. 6

END-TO-END SIGNAL FLOW Connected vs. Autonomous Operation

CONNECTED MODE

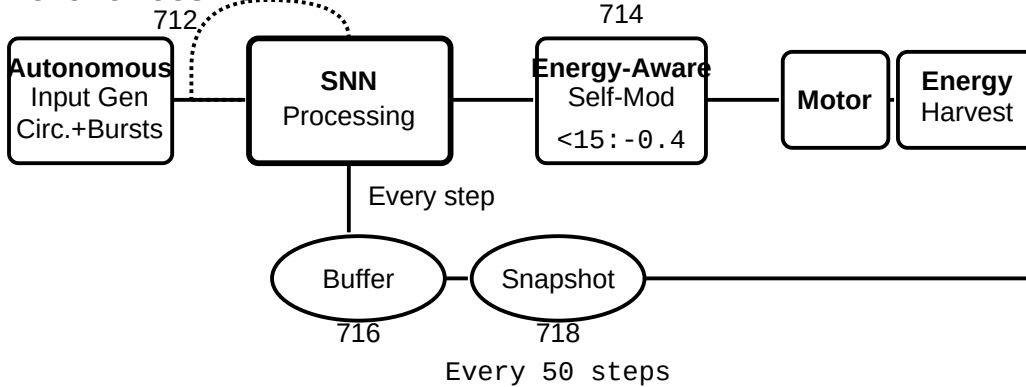


DISCONNECTION EVENT (heartbeat timeout > 30s)
Seamless transition - no interruption to SNN or Harvester

720
resync()

Self-Observation (Patent B)

AUTONOMOUS MODE



Signal Pathways

- Forward signal path
- - - - - Energy/control feedback
- Self-Observation feedback (Patent B)

FIG. 7